

Yuhan Hu

+1 (919) 757-7661 | yhu27@ncsu.edu | linkedin.com/in/yuhan-hu

Summary

PhD candidate with 4+ years of research experience in: (1) **prognostics and reliability prediction**, applying **statistical learning, privacy-preserving federated learning, and stochastic modeling**; and (2) **multimodal AI**, with emphasis on **LLM/VLM reasoning, evaluation, and calibration**.

Education

Ph.D., Industrial and Systems Engineering **Expected 2026**

North Carolina State University

- **Achievements:** Anderson Scholar — recognized for leadership in industry-academia engagement

M.S., Statistics; M.S., Industrial and Systems Engineering; M.S., Textile Engineering **2021-2026**

North Carolina State University

- **Achievements:** Provost's Fellowship Scholar

B.S., Fashion Design and Engineering **2020**

Zhejiang Sci-Tech University

- **Achievements:** Merit Scholarship

Technical Skills

- **Machine Learning & Statistical Modeling:** Supervised Learning, Federated Learning, High-Dimensional and Streaming Data Analysis, Data Fusion, Prognostics, Survival Analysis, Stochastic Modeling, Reliability Modeling, Failure-Mode Diagnosis, Experimental Design
- **Generative AI:** Multimodal Learning, LLM/VLM Reasoning, Prompt Engineering
- **Programming & ML Frameworks:** Python, R, MATLAB, PyTorch, TensorFlow, LangChain, Spark
- **Data Science & Analytics:** SQL, pandas, NumPy, Tableau

Research Experience

North Carolina State University — Department of Industrial and Systems Engineering **Raleigh, NC**

Research Assistant

Aug 2021 – May 2025

Personalized Federated Failure-Time Prediction

- Built a **privacy-preserving failure-time prediction model** for **heterogeneous data** via log-location-scale regression.
- Designed a similarity-aware mechanism enabling **knowledge sharing** across similar clients **without sharing raw data**.
- Proposed a **proximal gradient based algorithm** for decentralized parameter estimation, reducing the median MAPE by **1.77%** compared to conventional federated learning methods.

Federated Data Fusion-Based Prognostic Model

- Designed an **end-to-end privacy-preserving prognostic pipeline** for **high-dimensional, outlier-contaminated** data, evaluated on datasets comprising **200+ engines with 21 sensor channels**.
- Proposed a **federated failure-mode identification algorithm** that groups machines by failure mode without data sharing.
- Developed a **federated robust PCA data fusion method** based on ADMM, with Incremental SVD enabling **decentralized low-rank estimation**.

Reliability Modeling with Damage Resistance

- Proposed a **two-stage reliability model** capturing the interplay of degradation, random shocks, and damage resistance.
- Derived a **closed-form reliability function**, reducing tire-lifetime prediction error from multiple years using existing approaches to **<0.05 years**.

Human-Machine System Reliability Modeling

- Proposed a **human-machine interaction system reliability model** that integrates situation awareness, machine degradation, and external shocks.
- Quantified **situation awareness** as a **time- and automation-dependent factor**, which is more realistic than constant human-error-rate models.

Duke University — Department of Computer Science

Graduate Researcher

Durham, NC

May 2025 – Present

Vision-Dependent Mathematical Reasoning Benchmark

- Developed a **diagnostic decoupling framework** on mathematical problems with visual context, which compares end-to-end, AI-description, and human-description pipelines to disentangle **visual perception errors from reasoning failures**.
- Identified **visual perception** as a primary VLM bottleneck, with accuracy decreasing by over **35%** from describable to non-describable problems.
- Constructed a **visual reasoning benchmark** spanning nine non-describable task categories and three difficulty levels to evaluate **spatial simulation and pixel-level reasoning** beyond text-based proxies.

UNC Greensboro — Department of Computer Science

Graduate Researcher

Remote

Aug 2025 – Present

Modality-Aware Calibration for Multimodal LLMs

- Proposed a **calibration framework for multimodal LLMs**, which aligns modality importance with model reliance to mitigate unreliable inference.
- Developed a **supervised multimodal encoder** using cross-modal attention and gated fusion to estimate the importance of heterogeneous modalities.
- Implemented an **inference-time prompting** that injects modality importance and reliance, improving accuracy by up to **13.1%** across **clinical text-image and physiological-signal tasks** without updating LLM parameters.

Selected Publications

- **Hu, Y., & Fang, X.** A federated learning approach for personalized prognostics in the presence of user-level heterogeneity. *Submitted to IJSE Transactions (under review)*.
- **Sultana A., Hu, Y., Sun Y., & Tong Q.** MIRACLE: A modality importance and reliance-aware calibration framework for robust multimodal LLM inference. *Submitted to AAAI (under review)*.
- **Hu, Y., Wang, J., Vartak, R., & Dhingra, B.** Your Reasoning Benchmark May Not Test Reasoning: A Benchmark for Vision-Dependent Mathematical Reasoning. *In preparation*.
- **Hu, Y., Zhu, M., & Lin, H.** A nonlinear Wiener process degradation model with damage resistance for reliability analysis. *Published in Annals of Operations Research*.
- **Hu, Y., Wang, R., Zhu, M., & Chen, K. B.** Modeling human-machine interaction system reliability with multiple dependent degradation processes and situation awareness. *Published in International Journal of Reliability, Quality and Safety Engineering*.
- **Hu, Y., & Zhu, M.** Reliability modeling for shock-degradation-dependent process considering damage resistance. *Published in Proceedings of the 27th ISSAT International Conference on Reliability and Quality in Design*.
- **Hu, Y., & Zhu, M.** System reliability models with random shocks and uncertainty: A state-of-the-art review. *Published in Predictive Analytics in System Reliability*.

Teaching Experience

North Carolina State University — Department of Industrial and Systems Engineering

Raleigh, NC

Instructor

Aug 2026

- Taught linear algebra to incoming PhD students in Industrial and Systems Engineering and Operations Research.
- Developed course materials and practical examples covering linear programming, optimization formulations, and solution methods.

Professional Services

President | INFORMS NCSU Student Chapter, North Carolina State University

Aug 2024 – May 2025

- Organized seminars, job-search talks, and programming workshops
- Received INFORMS Student Chapter Annual Award

Reviewer | *Reliability Engineering & System Safety, Engineering Economist, IEEE Transactions on Reliability*